

Nyx Haile

Quantitative Developer

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EDUCATION

Massachusetts Institute of Technology

2026

B.S. Computer Science, B.S. Mathematics, Minor in Economics

Cambridge, MA

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

EXPERIENCE

Partial

2025 – Present

Cofounder & CEO

London

- Co-founded Partial, a prediction-market exchange where traders take positions on how correlated assets move together.
- Built the Rust pricing engine with my cofounder; it prices two-asset trades over ~200k-cell probability grids in 50-150 ms on a 2-CPU server.
- Designed eleven new classes of tradable contract, from price paths and correlations to lead-lag and reported statistics, all under one pricing framework.

Weiss Laboratory & The SHED, MIT

June 2025 – Sept 2025

Undergraduate Researcher (UROP), generative AI for vascularised liver tissue

Cambridge, MA

- Wrote a parametric vascular-network generator specification covering stochastic branching, tapering, R-tree collision avoidance and a 0.99 diffusion-coverage loop, sized to a 100x68x38 mm bioprinter envelope.

SELECTED PROJECTS

Repositories marked private are available on request.

MIT Pokerbots | Python

Heads-up poker-variant bot, January 2026

- Built a Python bot for MIT Pokerbots' heads-up poker variant; the team placed about 15th of 70.
- Implemented Monte Carlo equity estimation, with exact enumeration for discards, using quick and deep sampling paths to fit a per-decision compute budget.
- Built a multi-policy architecture (tight, bluff, desperate, lock-win) with a strategic overseer choosing among threshold policies.

howtowin.lol | Python, PyTorch, CUDA, Redis

World model for post-game teaching

- Designed a Dreamer-lineage RSSM world model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.
- Trained the world model end to end on a 12.8k-game corpus, reaching outcome AUC@15 of 0.88 game-cold and 0.84 player-cold with a clean leak probe.
- Designed leakage-controlled evaluation over 51,000 games: a game-cold holdout split by match-ID hash and a player-cold holdout of every match played by 594 hash-selected players.

Strand (private) | Rust, FoundationDB

Capability-authenticated issue tracker

- Wrote stable-Rust fuzz harnesses for three frozen v1 formats (wire frames, CBOR citations, backup files); the first pass ran 284,846 iterations with zero crashes.

Neuro (private) | Python, NumPy

Exploratory adaptive sparse neural core

- Implemented temperature-controlled sampled activation, copula-driven rewiring, a loss-plus-sparsity objective and automatic pruning with edge rebalancing.
- Wrote a research note relating the model to resource-bounded, stream-based approximations of Solomonoff induction.

Forge (private) | Rust, systemd, RPC

Local-first agent control plane

- Added bounded, jittered exponential backoff to scheduler dispatch, seeded per task so retry waves de-correlate in production and replay exactly in tests.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis